FLEET TEAM DIGEST

737MAX-FTD-31-21003

Issue Title : High Rate of "ALT DISAGREE" Flight Deck Effect and Correlated Maintenance Message 31-03130

Airplane Model	ATA	Minor Model(s)
737MAX	3100-00	-7, -8, -8200, -9, -BBJ
Other Model(s)	**ECCN**	
	9E991	

Originated Date	Last Revised Date	Created On
04/27/2021	02/28/2024	01/11/2024
Estimated Completion Date	**Next Update Date**	
12/14/2024	12/14/2024	

Revision Description

No update.

Applicability

737MAX airplanes installed with software "46 NFS OMF OSS" P/N BCG3B-0MFC-0005 or later.

Description

Boeing and operators have been observing a high rate of occurrence of Flight Deck Effect (FDE) **"ALT DISAGREE shows on the PFD – captain's"** (342 301 31) and **"ALT DISAGREE shows on the PFD – first officer's"** (342 301 32) with correlated Maintenance Message (MMsg) 31-03130 **"ALT DISAGREE SHOWS ON CAPT OR F/O PFD"** logged in Onboard Maintenance Function (OMF) on newly delivered 737MAX airplanes with software "46 NFS OMF OSS" P/N BCG3B-0MFC-0005 installed.

This FDE occurs when the altitude of the captain and the first officer display differ by more than 200 feet (60 meters) for five (5) seconds. Please see the attachment "ALT DISAGREE on PFD.pdf" for a visualization.

This FDE can be driven by multiple root causes. However, Boeing has observed that the large majority of the occurrences have been crew induced, and occurring during climb or descent, as given in the "Background" section.

Background

At transition altitude, flight crews change barometric altimeter settings to either standard (when climbing through transition altitude), or to local reference altimeter setting (when descending through transition altitude). Often, one crew member changes their barometric setting at a different time than the other crew member due to one crew member performing other tasks. If the

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/06/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

change in altimeter setting is greater than approximately 0.2 inHg (Inches of Mercury or 6.8 hPa) and the crew action is more than five (5) seconds apart, this will trigger the **ALT DISAGREE** FDE, which is expected behavior. In almost all cases, the crew resolves the difference quickly by changing the barometric settings to match, making the FDE inactive. The fault will not latch, an inactive **ALT DISAGREE** FDE will not set the maintenance light, and if induced in this manner, it is very unlikely to even be logged by the flight crew.

However, the FDE is being observed in Present Leg Fault (PLF) reports and Real Time Event (RTE) reports that are generated by OMF and transmitted via Aircraft Communications Addressing and Reporting System (ACARS.) In the crew induced fault cases, these usually display as inactive faults, with a duration of one minute or less, and in climb or descent.

The transmission of these faults have been limited to newer delivered 737MAX airplanes due to a change in the OMF logic. 46 NFS OMF OSS Loadable Software Airplane Part (LSAP) part number BCG3B-0MFC-0005, which is present on most newly delivered airplanes, added the MMsg 31-03130 **"ALT DISAGREE SHOWS ON CAPT OR F/O PFD"**. This was added based on airline feedback that a correlated MMsg enables easier maintenance. OMF correlates this MMsg with the FDE after being active on the displays for three (3) seconds. The FDE has a four (4) second persistence, therefore an active FDE will always be logged in OMF. Prior 46 NFS OMF OSS LSAPs (BCG3A-0MFC-0004 and prior) do not have this correlated MMsg. OMF logic only transmits faults with correlated MMsgs in PLF and RTE reports. This means that although the behavior in the flight deck is identical across all MAX aircraft, the FDE only appears in PLF and RTE reports for aircraft with part number BCG3B-0MFC-0005 or later of 46 NFS OMF OSS LSAP.

In Airplane Health Management (AHM), the PLF and RTE ACARS messages from OMF are translated and formatted. The FDE **ALT DISAGREE**, correlated MMsgs, and occurrence timestamps and duration timestamps are available.

Operators can view the current OMF version loaded on their aircraft via AHM by viewing a Fault Summary and viewing the value under "CMC or LDI Part Number". It can also be viewed on wing via ONBD MAINT -> LINE MAINT -> 46 – Information Systems -> 46 NFS OMF OSS Part Number.

NOTE: Boeing also included correlated MMsgs for the other Air Data Disagree FDEs (PITCH, ROLL, IAS, and AOA) in 46 NFS OMF OSS part number BCG3B-0MFC-0005, but these are not regularly crew induced.

Status

Boeing recognizes crew induced, no maintenance action required faults are undesirable and can mean that non crew induced ALT DISAGREE faults are possibly ignored.

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/06/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

Boeing is investigating potential improvements.

Interim Action

Boeing recommends operators to advise the flight crews of the five (5) seconds threshold when setting the altimeter between captain and first officer to minimize false MMsg 31-03130.

Additionally, operators may choose to set this fault as nuisance in AHM in order to reduce the review workload of Actionable Items, however operators should be warned this can inhibit non crew induced ALT DISAGREE FDEs from appearing as Actionable Items in AHM. Note, setting faults as nuisance in AHM requires AHM manager privileges.

To set a fault as nuisance in AHM, use the “Customize Priorities” selection under AHM Resources. Use code “3423013” (Corresponding to first 7 numbers of the two FDEs) in the search bar. Under the “Airline Priority” column, set the message to “Nuisance”. This will prompt for certain flight phases and activity. Boeing suggests “Climb” and “Descent” as flight phases and “Inactive” as activity. This will set these messages as non-actionable. They can still be viewed under History Items.

Final Action

Boeing is still investigating any potential Final Action.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Root Cause Established: 1Q2021 (Complete)

Solution Selected: TBD

Change Committed: TBD

Production Incorporation: TBD

Service Bulletin Available: TBD

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- Best Practices / Operational Efficiency : Operational / Maintenance-focused best practices, techniques, health checks or recommendations for preserving aircraft efficiency.

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/06/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

FLEET TEAM DIGEST

- Flight Deck Effects and Maintenance Messages : Issue that occurs frequently and causes a flight deck effect or maintenance message
- Flight Operations : Flight Operations

Attachments:

a) ALT DISAGREE on PFD

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/06/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991